AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2023

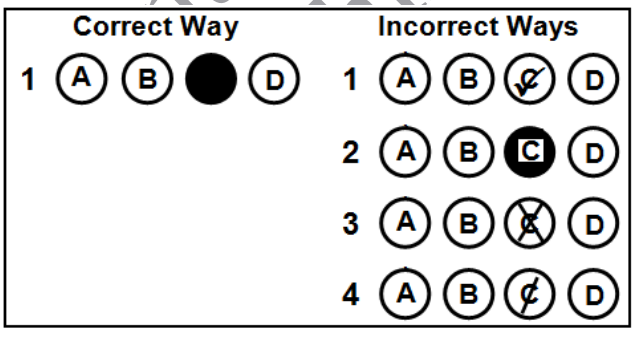
Business Statistics Paper I

Time: 55 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols have their usual meanings.

Measures of Central Tendency and Quartiles

$$\overline{X} = \frac{\sum wx}{\sum w} \qquad \overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{\sum f}$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Measures of Dispersion

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{R} = x_{\max} - x_{\min}$$

$$\text{variance} = \frac{\sum \left(x - \overline{X}\right)^2}{n} \qquad \text{SD} = \sqrt{\frac{\sum \left(x - \overline{X}\right)^2}{n}}$$

$$\text{variance} = \frac{\Sigma\, fx^2}{\Sigma\, f} - \left(\frac{\Sigma\, fx}{\Sigma\, f}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Counting Techniques and Probability

$${}^{n}P_r = \frac{n!}{(n-r)!} \qquad {}^{n}C_r = \frac{n!}{(n-r)!r!}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(A) \times P(B) \qquad P(A \cap B) = P(B) \times P(A \mid B)$$

Scatter Diagram and Correlation

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n \sum X^2 - (\sum X)^2\right]\left[n \sum Y^2 - (\sum Y)^2\right]}}$$

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1. In comparison to secondary data, all of the given are characteristics of primary data EXCEPT that it is

   A. cost effective.
   B. in crude form.
   C. time consuming.
   D. specific to need of research.

2. The questions for an effective questionnaire should be

   I. short
   II. lengthy
   III. asked randomly
   IV. relevant

   A. I and III.
   B. I and IV.
   C. II and III.
   D. II and IV.

3. In a shop, the sizes of shoes available are listed as 6, 7, 8, 9, 10, 11 and 12. This data set is an example of

   I. qualitative data
   II. quantitative data
   III. discrete data
   IV. continuous data

   A. I and III.
   B. I and IV.
   C. II and III.
   D. II and IV.

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Use the given information to answer Q.4 and Q.5.

The given frequency distribution represents the information about the number of pets owned by students in a class.

Number of Pets	Number of Students
1 - 2	7
3 - 4	9
5 - 6	5
7 - 8	1

4. The upper class boundary of the class interval with the highest frequency is

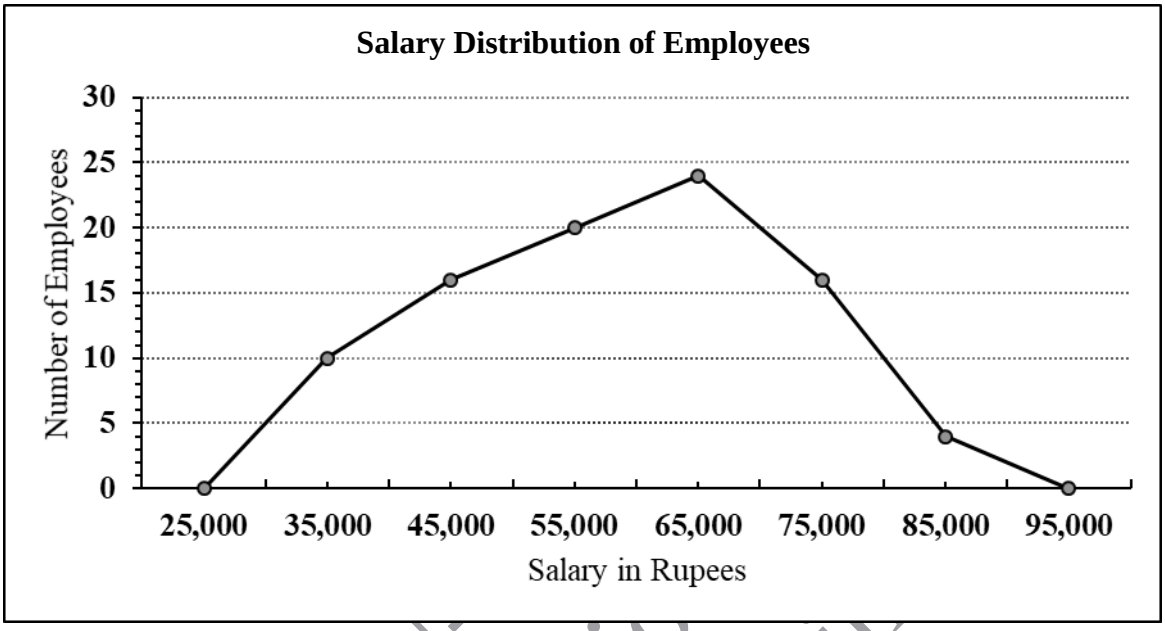
   A. 3.5
   B. 4.5
   C. 5.5
   D. 6.5

5. The class mark of the class interval with the least frequency is

   A. 1
   B. 5
   C. 6.5
   D. 7.5

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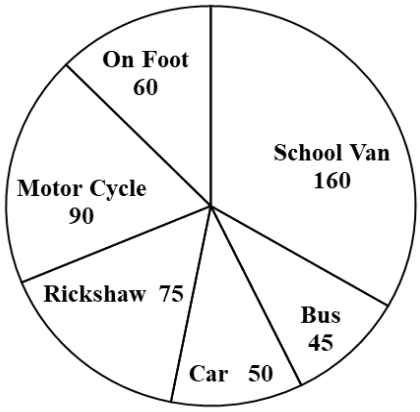
6. The given frequency polygon represents the salary distribution of the employees in an organisation.



In this organisation, the most number of employees belong to the class interval

A. 55,000 - 59,000
B. 60,000 - 70,000
C. 65,000 - 75,000
D. 80,000 - 90,000

7. For a school, the given pie chart represents the number of students and their modes of travelling to their school.



The central angle of the sector representing the number of students which DO NOT travel by any of the vehicle is

A. 12.5^{o}
B. 45°
C. 16.6^{o}
D. 80°

8. The given table shows the age-wise distribution of members of a sports club.

Age (Years)	Count of Members
5 - 10	40
10 - 15	75
15 - 20	100
20 - 25	45
25 - 30	60

In percentage, the relative frequency of the class interval with the maximum frequency is

A. 115
B. 100
C. 35.94
D. 31.25

9. A survey is conducted in a local community about the usage of smartphones. The results are compiled in the given frequency distribution.

Age of Participant (Years)	Usage of Smart Phone (Hours)
10 - 14	3
15 - 19	6
20 - 24	8
25 - 29	7
30 - 34	4
Total	28

The cumulative frequency of the class that is preceding the median class is

A. 5
B. 8
C. 9
D. 11

10. The number of goals scored in 10 football matches were 5, 4, 0, 5, 5, 0, 4, 0, 1 and 5.

The mean number of goals is

A. 2.7
B. 2.8
C. 2.9
D. 3.5

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11. The arithmetic mean of a set of values is 5. If each value is multiplied by 2 and two is subtracted from each of the value, then the new mean will be

    A. 3
    B. 5
    C. 8
    D. 10

12. The median of 2, 4, 1, 5, 3 and 5 is

    A. 3.5
    B. 4.0
    C. 5.0
    D. 5.5

13. The first quartile of the data 5, 6, 7, 8, 8, 9, 10 and 12 will be

    A. 5.5
    B. 6.0
    C. 6.5
    D. 7.0

Use the given information to answer Q.14. and Q.15.

Consider the given stem and leaf plot of a certain data.

1	5 6 7 8 8 9
2	0 1 2 4
3	2 9
4	5 6

14. Two values in the data are equal. The equal values are

    A. 8 and 8
    B. 6 and 6
    C. 9 and 9
    D. 18 and 18

15. The median of the data is

    A. 20.0
    B. 20.5
    C. 21.0
    D. 21.5

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16. The variance in the data $2x+a, 2x+a, 2x+a, 2x+a, 2x+a$ and $2x+a$ is

A. 0
B. $4x$
C. a^2
D. $(2x+a)^2$

17. For two sets of data, the information is presented in the following table.

	Sum of Data $\sum X$	**Mean of Data** $\overline{X}$	**Standard Deviation of Data** $S.D$
Data Set I	20	4	1.095
Data Set II	25	5	0.632

The number of elements in the data set I and the data set II respectively are

A. 4 and 4
B. 4 and 5
C. 5 and 4
D. 5 and 5

18. The variance in the values $1, 2, -2$ and 3 is 3.5.

If each of these values is reduced to half, then the standard deviation will be

A. 0.935
B. 1.750
C. 1.871
D. 3.500

19. If the variance of variable X is 39.5, then the variance of the variable $4X$ will be

A. 9.875
B. 39.5
C. 158
D. 632

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20. Consider the given table representing the price of four commodities.

Commodity	Price on March 30, 2022	Price on June 30, 2022
P	110	125
Q	98	105
R	60	80
S	52	60

By taking the price on March 30 as a base, the index number reflects that prices of all four commodities on June 30 are

A. 15.625% higher than the base price.
B. 15.625% lesser than the base price.
C. 86.486% lesser than the base price.
D. 86.486% higher than the base price.

21. Consider the given table of values.

Commodity	Price per kg (P_o) 2015	Quantity kg (Q_o) 2015	Price per kg (P_n) 2020	Quantity kg (Q_n) 2020	$P_o Q_o$	$P_n Q_o$
Rice	120	15	150	12	1,800	2,250
Sugar	75	5	110	6	375	550
Flour	60	20	90	15	1,200	1,800
Milk	110	25	130	20	2,750	3,250
Total	365	65	480	53	6,125	7,850

For the given data, the laspeyres price index is

A. 131.51
B. 128.16
C. 78.03
D. 76.04

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22. The factorial expression $\frac{0!n!}{(n-1)!}$ is equal to

A. 0
B. n
C. $\frac{1}{n}$
D. $\frac{1}{(n-1)}$

23. A bookshelf has 15 books. The number of ways in which any 3 books can be randomly chosen from this shelve is

A. 45
B. 220
C. 455
D. 2730

24. The number of different four-lettered arrangements of letters of the word 'BLUE', without repetition, is

A. 4
B. 12
C. 24
D. 256

25. In a weight-lifting training camp, 8 players took part. If a team of 4 players is to be selected from these players, then the possible number of team selections will be

A. 70
B. 256
C 1,680
D. 4,096

26. In a stack of nine books, six books are of science fiction and three books are of suspense.

The different number of ways to select three books such that two books are of science fiction and one book of suspense is

A. $\frac{9!}{6!3!}$
B. $6^2 \times 3$
C. $\frac{6!}{4!} \times 3!$
D. $\frac{6!}{2!4!} \times \frac{3!}{2!}$

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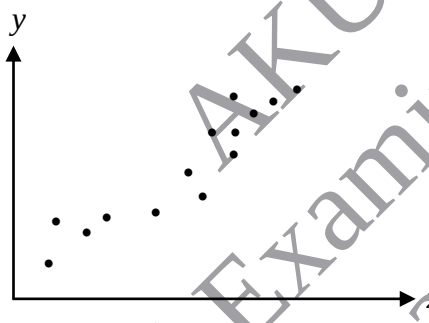
27. If $P(A)=P(B)=a$ and $P(A\cap B)=b,$ then $P(A\cup B)$ is equal to

A. $a+b$.
B. $a-b$.
C. $2a+b$.
D. $2a-b$.

28. Two fair dice are rolled together. The probability that both dice show an even number is

A. $\frac{9}{36}$
B. $\frac{12}{36}$
C. $\frac{15}{36}$
D. $\frac{18}{36}$

29. For the given scatter diagram, the linear correlation between x and y could be



A. −1
B. 0.5
C. 0.8
D. 1

30. If the graph of two variables X and Y represents a horizontal line, then the coefficient of correlation between these two variables is

A. −1
B. 0
C. 0.5
D. 1

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END OF PAPER

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